

Linear Algebra

* 1.

This problem involves 4×4 matrices with entries in $\mathbb{R}$. We will say that a matrix M is an ***E*-matrix** if M has 1's along the diagonal and a single nonzero entry off the diagonal. For instance:

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 31 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

is an *E*-matrix. Express the matrix A below as a product of *E*-matrices.

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \end{pmatrix}$$

Solution 1.

This is a straight up row reducing problem. Let $E_{i,j}$ be the matrix with 1s on the diagonal, a 1 in position (i, j) , and 0s elsewhere. Then

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \end{pmatrix} = E_{2,1}E_{3,1}E_{4,1}E_{1,2}E_{1,3}E_{1,4}$$

2.

- (a) Give an example of a finite dimensional vector space V over $\mathbb{C}$, and a nonzero linear transformation $A: V \rightarrow V$ whose only eigenvalue is 0.
- (b) Prove that the only eigenvalue of $A: V \rightarrow V$ is 0 if and only if A is nilpotent. (V is still f.d. over $\mathbb{C}$.)

Solution 2.

- (a) Let $V = \mathbb{R}^2$, and A be the linear transformation corresponding to the matrix

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

- (b) If A is nilpotent, then it satisfies the polynomial $x^k = 0$ for some k . But then the minimal polynomial of A divides x^k , and so its only eigenvalue is 0.

On the other hand, suppose the only eigenvalue of A is 0. Then the characteristic polynomial of A is a degree n polynomial whose only root is 0, which means it must be x^n . By the Cayley-Hamilton Theorem, A satisfies its own characteristic polynomial, so $A^n = 0$.

3.

Give an example of a nonzero finite-dimensional vector space V over $\mathbb{C}$ and an invertible linear transformation $A: V \rightarrow V$ such that A is not diagonalizable.

Solution 3.

The classic example is the shear transformation:

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

One way to think about Jordan normal form is that it shows that shearing is *exactly* the obstruction to being diagonalizable.

*** 4.**

Let M be an $n \times n$ matrix with real entries. Let M^T be its transpose. Each of these two matrices defines a linear transformation $\mathbb{R}^n \rightarrow \mathbb{R}^n$; let A denote the transformation corresponding to M , and let B denote that corresponding to M^T .

- (a) Prove that any vector in $\ker(A)$ is orthogonal to any vector in $\operatorname{im}(B)$ with respect to the standard inner product.
- (b) Prove that $\mathbb{R}^n$ is the direct sum of $\ker(A)$ and $\operatorname{im}(B)$.

Solution 4.

- (a) Let $v \in \ker(A)$ and $u' \in \operatorname{im}(B)$. Then there is some $u \in \mathbb{R}^n$ so that $u' = M^T u$. Then we have $v \cdot u' = v \cdot (M^T u) = Mv \cdot u = 0 \cdot u = 0$.
- (b) By the rank-nullity theorem,

$$n = \operatorname{rank}(M) + \operatorname{null}(M) = \operatorname{rank}(M^T) + \operatorname{null}(M) = \dim \operatorname{im}(B) + \dim \ker(A).$$

Also, since any vector in $\ker(A)$ is orthogonal to any vector in $\operatorname{im}(B)$, $\ker(A) \cap \operatorname{im}(B) = \{0\}$. Thus, $\mathbb{R}^n = \ker(A) + \operatorname{im}(B)$.

*** 5.**

- (a) Let V be a finite-dimensional vector space over $\mathbb{C}$, and let $T: V \rightarrow V$ be a linear transformation. Show that T has an eigenvector.
- (b) Give an example of a nonzero finite-dimensional vector space V over $\mathbb{R}$ and a linear transformation $T: V \rightarrow V$ such that T does not have an eigenvector.
- (c) Does a linear transformation of an infinite-dimensional vector space over $\mathbb{C}$ have to have an eigenvector? Either prove this is the case, or give an example of a linear transformation of an infinite-dimensional vector space with no eigenvector.

Solution 5.

- (a) Let $\chi(x)$ be the characteristic polynomial of T . Then by the fundamental theorem of algebra, χ has a root λ . This means that $\det(T - \lambda \operatorname{Id}) = 0$, so $T - \lambda \operatorname{Id}$ has nontrivial kernel. For any $v \in \ker(T - \lambda \operatorname{Id})$, we have $Tv = \lambda v$, so v is an eigenvector for T .

- (b) The classic example is a rotation by 90° :

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

- (c) This does not have to be the case. Consider the vector space V whose elements are sequences of complex numbers with only finitely many nonzero terms. So a vector looks like $v = (z_1, z_2, \dots, z_t, 0, 0, 0, \dots)$, where t might be different for different vectors v , but is always finite. Let T be the "right shift" operator, so $Tv = (0, z_1, z_2, \dots, z_t, 0, \dots)$. If we try setting up $Tv = \lambda v$, then looking at the first entry tells us the only possible value of λ is 0, but T has no kernel so it can't have any eigenvectors.

6.

Let V and W be finite dimensional vector spaces over the same field (for the sake of argument, let's say over $\mathbb{R}$). Let $T: V \rightarrow V$ and $U: W \rightarrow W$ be linear maps. Consider the linear map $T \oplus U: V \oplus W \rightarrow V \oplus W$ which takes (v, w) to (Tv, Uw) .

- (a) Let $\mathcal{B}_1 = \{v_1, \dots, v_n\}$ be a basis for V , and $\mathcal{B}_2 = \{w_1, \dots, w_m\}$ be a basis for W . If A is the matrix of T with respect to $\mathcal{B}_1$, and B is the matrix of U with respect to $\mathcal{B}_2$, what is the matrix of $T \oplus U$ with respect to the basis $\mathcal{B} = \{(v_1, 0), \dots, (v_n, 0), (0, w_1), \dots, (0, w_m)\}$?
- (b) What are the eigenvalues of $T \oplus U$ in terms of the eigenvalues of T and the eigenvalues of U ?
- (c) What is the trace of $T \oplus U$ in terms of the trace of T and the trace of U ?
- (d) Same question for the determinant of $T \oplus U$.
- (e) Same question for the characteristic polynomial of $T \oplus U$.
- (f) Let $\mu_T(x)$ be the minimal polynomial of T , and $\mu_U(x)$ be the minimal polynomial of U . Prove that the minimal polynomial of $T \oplus U$ is $\text{lcm}(\mu_T, \mu_U)$.

Solution 6.

- (a) The matrix of $T \oplus U$ with respect to $\mathcal{B}$ is the block diagonal matrix

$$\begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}.$$

- (b) If $\lambda_1, \dots, \lambda_n$ are the eigenvalues of T , and $\lambda_{n+1}, \dots, \lambda_{n+m}$ are the eigenvalues of U , then $\lambda_1, \dots, \lambda_{n+m}$ are the eigenvalues of $T \oplus U$.
- (c) $\text{Tr}(T \oplus U) = \text{Tr}(T) + \text{Tr}(U)$
- (d) $\det(T \oplus U) = \det(T) \det(U)$
- (e) $\chi_{T \oplus U}(x) = \chi_T(x) \chi_U(x)$

- (f) Let $f = \text{lcm}(\mu_T, \mu_U)$. Since both $\mu_T \mid f$ and $\mu_U \mid f$, we certainly have that $f(T \oplus U) = 0$. Now, let g be any other polynomial such that $g(T \oplus U) = 0$, and we will show that $f \mid g$. Since $g(T \oplus U) = 0$, then for any $v \in V$, $g(T \oplus U)(v, 0) = (g(T)v, g(U)0) = (0, 0)$, so $g(T) = 0$, and thus $\mu_T \mid g$. Similarly, $\mu_U \mid g$, and hence $f \mid g$.

* 7.

Suppose V is a finite dimensional vector space over $\mathbb{C}$, and $T: V \rightarrow V$ is a linear operator. Suppose the subspace of V spanned by all eigenvectors of T is 1-dimensional. Prove that the minimal polynomial of T is $(x - \alpha)^n$, where $n = \dim V$ and α is some complex number.

Solution 7.

Not using Jordan normal form: the characteristic polynomial of T is a degree n polynomial whose only root is α , so it must be $(x - \alpha)^n$. By the Cayley-Hamilton theorem, T satisfies its minimal polynomial, so the minimal polynomial of T divides $(x - \alpha)^n$. We just need to check that $(T - \alpha \text{Id})^k \neq 0$ for any $k < n$. Notice that $\dim \ker(T - \alpha \text{Id}) = 1$, and for any linear transformations A, B , $\dim \ker(AB) \leq \dim \ker(A) + \dim \ker(B)$. Thus, $\dim \ker((T - \alpha \text{Id})^k) \leq k < n$, so $(T - \alpha \text{Id})^k \neq 0$.

Using Jordan normal form: if we take the Jordan normal form of T , all the Jordan blocks have α down the diagonal. However, if there is more than one block, then T will have two linearly independent eigenvectors, so actually the Jordan form of T is

$$\begin{pmatrix} \alpha & 1 & 0 & \dots & 0 \\ 0 & \alpha & 1 & \dots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \dots & \dots & \alpha & 1 \\ 0 & \dots & \dots & 0 & \alpha \end{pmatrix}.$$

From here, one can compute that $(T - \alpha \text{Id})^n = 0$, but $(T - \alpha \text{Id})^k \neq 0$ for any $k < n$.

8.

- (a) Give an example of two linear transformations with the same characteristic polynomial but different minimal polynomials.
- (b) Give an example of two linear transformations with the same minimal polynomials which are not similar.

Solution 8.

- (a) Consider

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

- (b) Here's where 6 (f) comes in handy. Consider

$$\begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

* 9.

(Originally part (d) of problem 5.) Suppose that T and U are two linear transformations on a finite-dimensional vector space V over $\mathbb{C}$ which satisfy $TU = UT$. Prove that there is some $v \in V$ which is an eigenvector for both T and U .

Solution 9.

Let λ be an eigenvalue for T , and let $W \subset V$ be the subspace consisting of all eigenvectors for λ (and the zero vector). (We know we can find such things by 5 (a).) Here's the "standard trick": for any $w \in W$, we have that $TUw = UTw = \lambda Uw$, so $Uw \in W$. Thus, we can think about U as a linear transformation on the smaller vector space W , and by 5 (a), U must have an eigenvector w' . But then w' is an eigenvector for both U and T , as desired.